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$$-60i + 11 = (x + yi)^2$$

$$-60i + 11 = x^2 + 2xyi - y^2$$

$$\begin{cases} x^2 - y^2 = 11 \\ 2xy = -60 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 11 \\ x^2y^2 = (-30)^2 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 11 \\ x^2(-y^2) = -900 \end{cases}$$

$$\lambda^2 - 11\lambda - 900 = 0$$

$$D = (-11)^2 - 4 \cdot (-900) = 3721$$

$$\lambda_1 = \frac{11 + \sqrt{3721}}{2} = 36 = x^2 \Rightarrow x = \pm 6$$

$$\lambda_1 = \frac{11 - \sqrt{3721}}{2} = -25 = -y^2 \Rightarrow y = \pm 5$$

$$x + yi \in \{6 - 5i, -6 + 5i\}$$

References